

Chargeback Fraud Detection on Anonymised Merchant Data: An Industry Case Study

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Abstract

Chargeback fraud — where legitimate transactions are later illegitimately disputed — imposes a significant cost burden on Australian merchants. Yet this area remains understudied in comparison to real-time fraud detection. We present an industry case study on detecting chargeback fraud using heavily anonymised transaction data from the Novatti Group Ltd, an Australian payments processing company. Our dataset comprises 126,184 transactions with extreme class imbalance (fraud rate of 0.31%), with customer PII, card identifiers, and IP addresses removed to comply with privacy regulations. Despite the loss of traditional fraud signals, our models achieve an ROC-AUC of 87%, demonstrating that effective detection remains possible on anonymised data. We observe an extreme concentration of fraud in this dataset, with 22% of merchants accounting for 80% of all fraud cases. Using this insight, we develop a privacy-preserving feature focused on merchant-level patterns rather than individual customer behaviour. We evaluate this merchant risk scoring feature against two leading class-imbalance techniques: SMOTE and Isolation Forests. The results show that the merchant risk scoring technique improves performance by encoding historical fraud patterns, whereas SMOTE degrades precision by a factor of 3.4. Isolation Forest fails ($F1=0.013$), revealing that anonymised fraud lacks the geometric isolation properties required for leading anomaly detection methods. Our results provide actionable insights for payment processors operating under privacy constraints.

CCS Concepts

• Security and privacy → Intrusion detection systems; • Computing methodologies → Machine learning.

Keywords

Fraud Detection, Anomaly Detection, Machine Learning, Imbalanced Learning

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1 Introduction

Australian card fraud reached a record \$913 million in 2024, with card-not-present transactions accounting for 89% of losses [4]. While considerable research addresses real-time fraud detection at the point of authorisation, a distinct problem class receives comparatively less attention: *chargeback fraud*, where transactions initially appear legitimate but are later illegitimately disputed by cardholders [1, 15]. Chargebacks represent a significant operational burden for merchants and payment processors, requiring investigation of disputed transactions weeks or months after settlement [17].

Detecting chargeback fraud presents unique challenges compared to real-time detection [20]. Traditional fraud signals, such as IP geolocation, device fingerprinting, behavioural biometrics, and velocity patterns, are designed for authorisation-time decisions. Whereas chargeback prediction must instead rely on transaction-level features that correlate with future dispute likelihood. Furthermore, privacy regulations increasingly require the anonymisation of sensitive data before analysis, thereby eliminating many features that practitioners consider essential for fraud detection [2, 3].

This paper addresses the *impact of feature anonymisation* on model performance, an area that remains understudied. Production systems routinely apply principal component analysis transformations, tokenisation, or encryption to meet compliance requirements, yet the performance implications of these operations are rarely quantified. We present an industry case study based on 126,184 transactions from Novatti Group Ltd., an Australian payment processor specialising in both domestic and cross-border transactions. The dataset exhibits extreme class imbalance with a fraud rate of 0.31%. We applied strict anonymisation, removing customer personally identifiable information, Bank Identification Numbers (BINs), IP addresses, and device identifiers. Of the 41 original features, only 21 remained usable after preprocessing, reflecting the realities of privacy-constrained environments.

Our analysis reveals that *fraud concentrates heavily at the merchant level*, with 7 of 32 merchants (22%) accounting for 80% of all fraud cases, with two outliers exhibiting fraud rates exceeding 90%. This concentration has important implications. First, it enables effective detection despite anonymisation, as merchant identity features contribute the most to the predictive power of our models (29.4% importance). Second, it suggests that targeted merchant-level interventions could address most of the fraud exposure without requiring individual customer surveillance. Third, it explains why generic class-imbalance techniques fail: SMOTE cannot synthesise realistic fraud patterns, and Isolation Forest cannot isolate fraud when distinguishing features have been anonymised.

We benchmark multiple model architectures, evaluate three class-imbalance techniques, and translate the results into dollar-denominated business impact. Our best model achieves an ROC-AUC of 87% with 86% precision at 47% recall, demonstrating that privacy-preserving fraud detection is achievable on highly anonymised chargeback data.

In summary, we make the following contributions:

- We present an industry case study demonstrating that effective chargeback fraud detection (87% ROC-AUC) remains achievable on heavily anonymised transaction data.
- We show that fraud concentrates at the merchant level, with 22% of merchants accounting for 80% of fraud cases, enabling privacy-preserving detection strategies.
- We evaluate common class-imbalance techniques, finding that SMOTE and Isolation Forest fail on anonymised data, whereas domain-specific merchant risk scoring succeeds.

2 Background and Related Work

Australian card fraud reached a record \$913 million in 2024, with card-not-present transactions accounting for 89% of losses [4]. While domestic fraud rates have declined following the implementation of AusPayNet’s Card-Not-Present Fraud Mitigation Framework, the scale of losses continues to motivate investment in detection capabilities.

Machine learning methods are the leading approach within industry and academia for fraud detection. For tabular data, gradient boosting methods [7, 13, 16] have proven effective, achieving state-of-the-art performance while maintaining speeds that enable real-time transaction scoring for small datasets. For sequential and time-series data, neural network-based approaches are preferred due to their ability to uncover spatio-temporal regularities [5, 8]. Further, graph neural networks have proven effective for detecting fraud rings by modelling entity relationships [9, 19]. The central limitation of these network-based approaches is the requirement for larger training datasets. Thus, traditional ML methods remain the standard for most small- to medium-sized organisations.

Within fraud detection, class imbalance remains the central methodological challenge, with fraud rates typically below 1%. Methods to address this challenge include data generation approaches, such as the synthetic oversampling method SMOTE [6], cost-sensitive learning [21], and AUC-oriented training [12]. Several reviews [11] provide comprehensive comparisons of ensemble methods for imbalanced learning.

Chargebacks constitute a distinct problem class, as disputes of authorised transactions are delayed, thereby requiring different signals than real-time fraud detection [18, 20]. Industry case-studies on the handling of chargeback-specific fraud detection remain sparse, and no systematic investigation of model performance has been conducted. To meet anonymisation requirements, production fraud systems routinely apply PCA transformations, tokenisation, or encryption, yet the performance implications of these methods remain understudied. Our work addresses this gap through a case study on heavily anonymised chargeback data.

3 Methods

3.1 Dataset Details

We analyse real-world transaction data from Novatti Group Ltd., comprising 126,184 merchant transactions collected between January 2023 and June 2025. Each record represents a transaction that received a chargeback claim, with the target variable indicating whether the chargeback was validated or rejected.

The dataset exhibits extreme class imbalance: only 394 transactions (0.31%) were confirmed fraud, while 125,790 (99.69%) were legitimate. This 319:1 imbalance ratio reflects the rarity of successful chargeback fraud in real-world payment systems and presents a significant modelling challenge.

3.2 Anonymisation and Feature Selection

To comply with privacy regulations, the dataset underwent strict anonymisation before analysis. Customer personally identifiable information (PII), including names, email addresses, physical addresses, dates of birth, and customer identifiers, was replaced with random strings. Similarly, sensitive card details, such as BINs, the last four digits of the card number, masked PANs, cardholder names, and issuing bank information, were anonymised. Network-level data, including source IP addresses and transaction references, was also removed.

This anonymisation eliminates many features traditionally used in fraud detection systems, such as IP geolocation, device fingerprinting, and card-level velocity checks. From the original 41 features, we retained 21 features after preprocessing, comprising:

- **Categorical features (10):** Payment method, card network (Visa/Mastercard), card type (credit/debit), card tier category, transaction status, merchant features, and product description.
- **Numeric features (11):** Transaction amount, surcharge, card expiry date components, settlement date features (day of week, month, weekend flag), and engineered features including log-transformed amount, months until card expiry, and high-value transaction indicator.

Merchant features are denoted MF-1, MF-2, and MF-3 in figures to protect industry partner information.

3.3 Data Preprocessing

Missing values in categorical features were imputed with an “UNKNOWN” category that is later encoded, while numeric features were imputed using median values. We engineered several derived features to capture potential fraud signals: transaction amount buckets, a binary flag for high-value transactions (> \$1,000), and the number of months remaining until card expiry (as domain experts within the partner company noted that cards approaching expiration may indicate stolen credentials).

The dataset was split into training (80%, $n=100,947$) and test (20%, $n=25,237$) sets using stratified sampling to preserve the 0.31% positive class ratio in both partitions. A fixed random seed ensured reproducibility.

3.4 Model Training and Evaluation

We employed AutoGluon [10], an automated machine learning framework, to benchmark multiple model architectures, including gradient boosting methods (LightGBM, XGBoost, CatBoost), tree ensembles (Random Forest, Extra Trees), and lightweight neural networks. AutoGluon was configured with the `medium_quality` preset, training 11 base models with automatic hyperparameter selection and a weighted ensemble.

Given the extreme class imbalance, we selected the F1 score as the primary evaluation metric, as it balances precision and recall for the minority fraud class. We also report ROC-AUC and Precision-Recall AUC (PR-AUC) as secondary metrics. Feature importance was computed using permutation importance on the held-out test set.

3.5 Addressing Class Imbalance

To address the extreme class imbalance, we evaluated three additional approaches beyond standard supervised learning:

Anomaly Detection (Isolation Forest): We treated fraud as an anomaly detection problem, using Isolation Forest [14] with contamination parameter set to the observed fraud rate (0.31%). Features were standardised prior to training.

Synthetic Oversampling (SMOTE): We applied the Synthetic Minority Over-sampling Technique [6] to generate synthetic fraud examples, balancing the training set to a 1:1 class ratio before training a Random Forest classifier with balanced class weights.

Merchant Risk Features: Motivated by the dominance of merchant-level features in our initial analysis, we engineered a merchant risk score using Bayesian-smoothed historical fraud rates. For each merchant, we computed:

$$\text{risk score} = \frac{n_{\text{fraud}} + \alpha \cdot r_{\text{global}}}{n_{\text{total}} + \alpha} \quad (1)$$

Where n_{fraud} and n_{total} are the merchant’s fraud count and total transactions, r_{global} is the global fraud rate, and $\alpha = 100$ is a smoothing factor that regularises estimates for merchants with few transactions. Similar risk features were computed for payment methods, but they had minimal impact on detection rates and were not used in the final benchmarking.

4 Results

4.1 Feature Importance Reveals Merchant Dominance

Figure 1 presents the top 10 features ranked by permutation importance. Cumulatively, merchant-related features dominate the ranking, with MF-1 accounting for 29.4% of predictive power, followed by payment method (11.6%) and MF-2 (11.5%). Together, merchant features account for over 50% of the model’s importance.

Card category (10.6%) ranks fourth, capturing distinctions between card tiers that correlate with different fraud profiles. Amount-related features provide a secondary signal, while customer-level features, which would typically dominate fraud models, are absent due to anonymisation. This absence, rather than degrading performance, reveals that *fraud concentrates at the merchant level*.

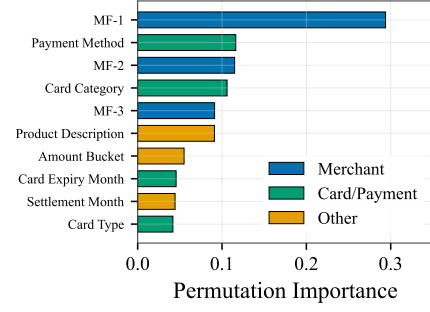


Figure 1: Top 10 features by permutation importance. Merchant features dominate and indicate fraud concentration in specific merchants.

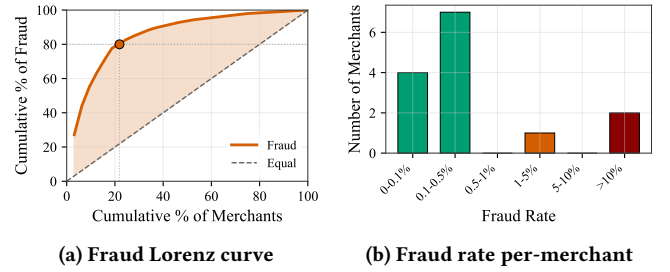


Figure 2: Merchant-level fraud concentration. (a) Lorenz curve indicating that 22% of merchants account for 80% of fraud, with the shaded region quantifying distributional inequality. (b) Heavy-tailed distribution of merchant fraud rates, with most merchants below 0.5% but outliers exceeding 10%, motivating the use for merchant-level features for detection models.

An insight we use to provide an alternate detection strategy, as discussed in Section 3.5.

4.2 The Merchant Concentration Pattern

Figure 2 quantifies the concentration of fraud at the merchant level. Seven merchants (22% of the 32 total) account for 80% of all fraud cases. The Lorenz curve in Figure 2a illustrates this inequality, with fraud events distributed far more unevenly than transactions.

Figure 2b reveals bimodal behaviour. Most merchants (25 of 32) have fraud rates below 1%, consistent with the base rate of 0.31%. However, two outliers exhibit fraud rates exceeding 90%, with almost every transaction from these merchants resulting in a fraudulent chargeback. These extreme cases likely represent compromised accounts or systematic fraud operations, presenting implications for merchant-level fraud mitigation. Targeted merchant interventions could address most of this fraud exposure without requiring individual customer surveillance.

Beyond merchant features, fraud correlates with transaction characteristics (Figure 3). The fraud rate for domestic transactions is more than 3× that for international transactions, at approximately 0.95% and 0.3%, respectively (Figure 3a). Credit card transactions have a fraud rate 8.7 times higher than debit card transactions, at

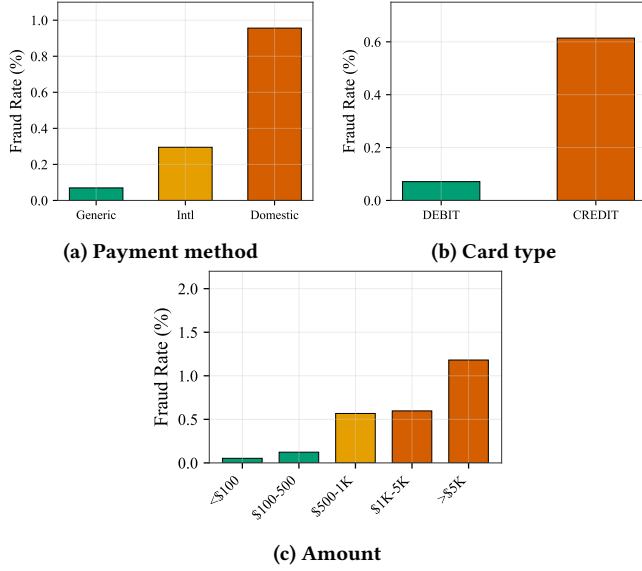


Figure 3: Fraud rate variation across transaction attributes. (a) Domestic payments exhibit the highest fraud rate among payment methods. (b) Credit cards exhibit a higher fraud rate than debit cards. (c) The rate of fraud increases with transaction amount, with transactions exceeding \$5K exhibiting the highest risk.

Table 1: Model performance on the test set, ranked by F1 score.

Model	F1	Precision	Recall	Time (s)
Neural Network	0.607	0.860	0.468	28.2
Random Forest	0.586	0.841	0.449	1.0
CatBoost	0.584	0.825	0.456	32.6
Extra Trees	0.579	0.831	0.443	0.5
LightGBM	0.579	0.820	0.443	1.0
XGBoost	0.537	0.782	0.418	0.3

0.61% and 0.07% respectively (Figure 3b), consistent with stronger consumer protections for the latter. Figure 3c shows that high-value transactions (>\$5,000) exhibit a 12× higher fraud rate than small transactions (<\$100), indicating that fraudsters target the maximum value per attempt.

4.3 Model Performance Despite Anonymisation

Given the merchant concentration pattern, we evaluate whether standard ML models can learn to detect fraud on anonymised data. Table 1 presents performance across architectures.

The neural network achieves the highest F1 score (0.607), with ROC-AUC of 87% and PR-AUC of 50%. At the default threshold, the model correctly identifies 37 of 79 fraud cases (47% recall) and generates only 6 false positives from 25,158 legitimate transactions (0.024% FPR). The precision score indicates that the model correctly flags fraud with 86% accuracy. These results demonstrate

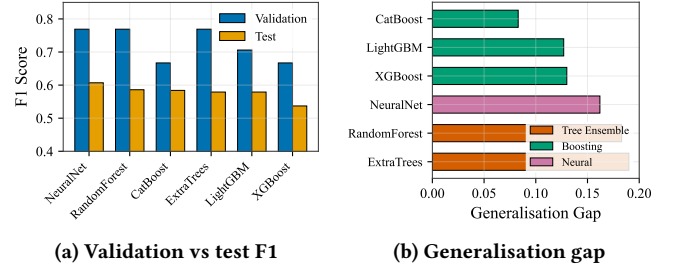


Figure 4: (a) Validation vs test F1 across architectures. (b) Generalisation gap by model: Tree ensembles exhibit larger gaps than boosting methods.

that effective fraud detection remains possible despite heavy data anonymisation, and the merchant concentration pattern provides a sufficient signal.

4.4 Generalisation and Error Analysis

Figure 4 reveals a consistent validation-test gap across all model architectures. Figure 4a compares the validation F1 scores (0.667 to 0.769) and the test F1 scores (0.537 to 0.607) for each model. In each instance, there is a noticeable drop in model performance between the validation and test sets. Figure 4b presents the generalised difference between the validation and test F1 scores for each architecture. Tree-based ensembles (Random Forest, Extra Trees) show the largest gaps (0.18), while gradient boosting methods (CatBoost, XGBoost) show smaller gaps (0.08 to 0.13). This validation-test gap pattern suggests overfitting to merchant-specific patterns in the training fold, which do not fully generalise to the held-out test set.

Error analysis provides deeper insight into model behaviour. Among the 43 false negatives (missed fraud), the average transaction amount (\$1,298) exceeds that for correctly detected fraud (\$915), suggesting that high-value fraud is more complex to detect. False negatives are dispersed across 14 merchants, whereas true positives are concentrated in just seven merchants. Demonstrating that the model excels at detecting fraud in high-risk merchants but struggles with fraud spread across the merchant population.

Conversely, the 15 false positives are concentrated in only four merchants and average \$610, similar to legitimate transaction amounts. These false alarms likely represent edge cases in merchants with mixed fraud profiles rather than systematic prediction errors. The low false-positive count (15 of 25,158 legitimate transactions, 0.06%) indicates that the model maintains high precision even when errors occur.

4.5 Analysis of Imbalance Handling

Addressing the large class imbalance of 319 (no fraud) to 1 (fraud), we evaluated three techniques designed for rare-class detection (Table 2).

4.5.1 Isolation Forest. Isolation Forest performs catastrophically, achieving an F1 Score of 0.013 and detecting only 1 of 79 fraud cases. The results reveal that fraud in anonymised data does not form isolatable clusters in feature space. This is because features

Table 2: Class imbalance technique comparison.

Method	F1	Prec.	Recall	AUC	FP
Baseline RF	0.554	0.706	0.456	0.921	15
Merchant Risk + RF	0.565	0.712	0.468	0.907	15
SMOTE + RF	0.452	0.427	0.481	0.915	51
Isolation Forest	0.013	0.013	0.013	0.808	74

that would create geometric separation — IP patterns, device fingerprints, and velocity anomalies — were precisely those removed during anonymisation. Fraud transactions look “normal” in the remaining feature space because the distinguishing behavioural signals are absent.

4.5.2 SMOTE. SMOTE increases recall marginally (from 45.6% to 48.1%) but reduces precision (from 70.6% to 42.7%), yielding 3.4× more false positives. Synthetic oversampling cannot create realistic fraud examples when the fraud pattern is *merchant concentration*. SMOTE interpolates feature values, but cannot synthesise transactions from high-risk merchants without understanding the underlying structure.

4.5.3 Merchant Risk Scoring. The use of merchant risk features and the derived merchant risk scoring feature improves performance by explicitly encoding the concentration pattern. This engineered feature becomes the most critical predictor (15.5% importance), improving F1 from 0.554 to 0.565 while maintaining the same low false-positive count. The approach succeeds because it directly encodes what the model was already implicitly learning: fraud is concentrated among specific merchants.

4.6 Dollar Cost Analysis

Figure 5 translates performance to dollars. Fraudulent transactions average \$1,107, compared with \$611 for legitimate transactions, indicating fraudsters target high-value purchases. Total fraud exposure in the test set is \$88,760. At the current classification threshold (47% recall, 15 FP), the model detects \$41,540 in fraud while \$47,220 goes undetected.

The precision-recall curve (Figure 6) reveals operational trade-offs: lowering the threshold to 90% recall reduces precision to 20%, increasing false positives by a factor of four. The break-even investigation cost is $\$47,220/15 = \$3,148$ per case. Therefore, if manual review costs less, lowering the threshold is justified; if it costs more, the current conservative configuration is optimal.

5 Discussion

Our results demonstrate that effective fraud detection models can be trained on highly anonymised transaction data, achieving an ROC-AUC of 87% despite the removal of traditional fraud signals, such as customer PII, card BINs, and IP addresses. This finding has practical implications for organisations seeking to balance fraud prevention with privacy compliance.

The extreme concentration of fraud suggests that targeted merchant-level interventions (e.g., enhanced monitoring, transaction limits, or account reviews) could address the fraud exposure with minimal operational overhead. Such an approach is inherently

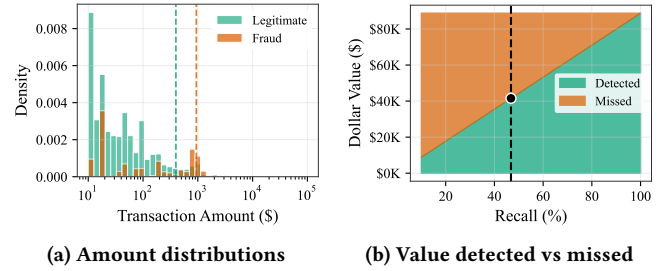


Figure 5: Financial perspective on fraud detection. (a) Fraudulent transactions exhibit systematically higher amounts than legitimate transactions (dashed lines indicate medians). (b) Dollar value detected versus missed across recall levels; at the operating point (approx. 42% recall, marker), the model recovers approximately \$40K in fraud, illustrating the direct financial impact of detection performance.

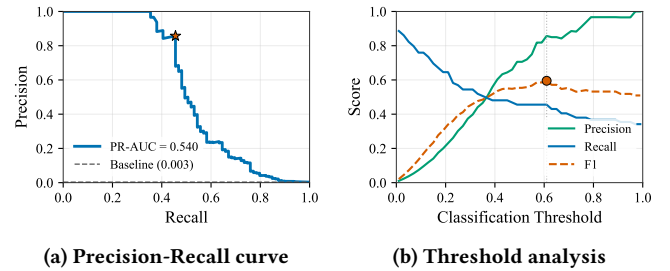


Figure 6: (a) Precision-recall curve demonstrating strong separation from random baseline (PR-AUC = 0.540 vs. 0.003), with optimal F1 of 0.607 marked. (b) Threshold sensitivity analysis showing a precision-recall trade-off; the optimal threshold balances false-positive reduction against fraud capture rate.

privacy-preserving for consumers, as it focuses on business entity patterns rather than on individual customer behaviour.

The observed precision-recall trade-off (86% precision, 47% recall) reflects a conservative model that prioritises avoiding false positives. In operational settings, this threshold can be adjusted to meet business requirements: organisations with high false-positive costs (e.g., customer friction) may prefer the current configuration, whereas those prioritising fraud-loss reduction may lower the threshold to capture more true positives at the cost of additional false alarms.

Our evaluation of class imbalance techniques yielded meaningful insights. Merchant risk scoring, which encodes historical fraud rates at the merchant level, provided the most significant improvement by transforming implicit merchant-level patterns into an explicit predictive feature. This approach is both interpretable and privacy-preserving. In contrast, SMOTE’s synthetic oversampling substantially degraded precision, suggesting that the generated samples do not faithfully represent the true fraud distribution. Most notably, Isolation Forest’s failure indicates that anonymised fraud does not exhibit the geometric isolation properties assumed by standard anomaly detection, as the distinguishing features (customer

behaviour, device fingerprints, velocity patterns) were precisely those removed during anonymisation.

Several limitations warrant consideration. First, the anonymisation process may have removed features that would substantially improve performance in a production system with full data access. Second, our analysis is limited to a single payment processor, and patterns may not generalise to different merchant verticals or geographic regions. Third, fraud patterns evolve over time, and model performance may degrade without periodic retraining on recent data.

6 Data and Code Availability

The dataset comprises real-world financial transactions provided under a confidentiality agreement with our industry partner. Due to privacy and regulatory constraints, public release is not permitted. Some implementation details are also withheld to prevent exploitation of any future deployed detection system. To support reproducibility within these constraints, we provide, where possible, procedures for feature engineering, model architectures, hyperparameter specifications, and dataset statistics. Our implementation uses standard open-source libraries; inquiries regarding methodological details are welcome.

7 Conclusion

We presented an industry case study on chargeback fraud detection using heavily anonymised merchant data from Novatti Group Ltd., an Australian payments processor. Despite removing customer PII, card identifiers, and network-level features, our models achieve an ROC-AUC of 87% and a precision of 86%. The key insight is that fraud concentrates at the merchant level: 22% of merchants account for 80% of fraud cases. This concentration enables privacy-preserving detection and explains why generic techniques like SMOTE and Isolation Forest fail on anonymised data. Merchant risk scoring, which explicitly encodes this concentration pattern, yields the most significant improvement. Our results demonstrate that effective fraud detection remains achievable under strict privacy constraints, with actionable implications for payment processors balancing security and compliance requirements.

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